

CTU Training Solutions

NET731 – Networking 3A

Week 2 Practical: Cloud Network Design

13 – 19 March 2025 | Total Marks: 10

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DECLARATION OF AUTHENTICITY

I, _____ Tristan Piek _____

(FULL NAME)

hereby declare that the contents of this assessment, including any contributions from group members, _____ NET Week 2 Practical _____

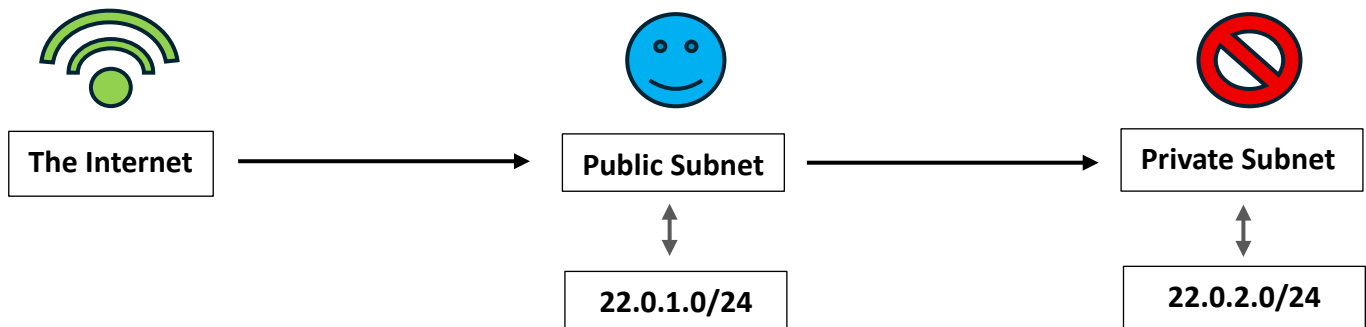
are entirely our own work, completed by the group without any paraphrasing, copying, or presenting as our own work accessed from any AI Apps, such as ChatGPT, Co-pilot, Perplexity, or any other App.

I have read and understood the CTU policies regarding referencing, cheating and plagiarism.

Signature(s): _____  _____

Date: _____ 08/05/2026 _____

Network Topology Diagram



2. IP Address Plan

The Vnet Name was chosen to be TechBridge-Solutions-SA-Vnet and the address was chosen as 22.0.0.0/18

3. Subnet Plan

The two subnets were created and was named

1. Public-Subnet
2. Private Subnet

And there CIDR notation is

1. 22.0.1.0/24
2. 22.0.2.0/24

NSG Rule Table

Rule Name	Protocol	Port	Direction	Action	Justification
Enable HTTP	TCP	80	All Inbound	Allow	Allows Web traffic inbound
Enable HTTPS	TCP	443	-	Allow	Allows secure traffic inbound
Deny all Inbound	ANY	*	-	Deny	Blocks unauthorized incoming traffic

Challenges Encountered

While doing week 2, I encountered two challenges. Firstly, when uploading the screenshots to the GitHub Repo I struggled with the naming convention of the images, but after researching how to properly insert screenshots I corrected my previous errors. Secondly, I encountered troubles while setting the NSG rule priorities. I researched proper rule testing procedures and how the rules must be structured and successfully created the NSG rules.

Screenshots (Just for Encase)

This screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation links for 'Create a resource', 'Home', 'Dashboard', 'All services', and a 'FAVORITES' section with links to 'Application Templates', 'All resources', 'Resource groups', 'App Services', 'Function App', 'Azure SQL Database', 'Azure Cosmos DB', 'Virtual machines', 'Load balancers', 'Storage accounts', 'Virtual networks', 'Microsoft Entra ID', 'Monitor', 'Advisor', 'Microsoft Defender for Cloud', and 'Cost Management + Billing'. The main content area is titled 'Home > Resource Manager | Resource groups' and displays the 'TechBridge-Solutions-SA-TP' resource group. Below the title, there's a search bar and a list of actions: '+ Create', 'Manage view', 'Delete resource group', 'Refresh', 'Export to CSV', 'Open query', 'Assign tags', and 'Group by none'. The 'Overview' tab is selected, showing a list of resources. The 'Resources' section has a filter bar with 'Filter for any field...', 'Type equals all', and 'Location equals all'. The resources listed are:

Name	Type	Location
TechBridge-Solutions-NSG	Network security...	South Africa N...
TechBridge-Solutions-SA-VNet	Virtual network	South Africa N...

At the bottom, it says 'Showing 1 - 2 of 2. Display count: auto' and has a 'Give feedback' link.

This screenshot shows the Microsoft Azure portal interface for the 'TechBridge-Solutions-SA-VNet' virtual network. The left sidebar is the same as the previous screenshot. The main content area is titled 'Home > Network foundation | Virtual networks' and displays the 'TechBridge-Solutions-SA-VNet' virtual network. Below the title, there's a search bar and a list of actions: 'Move', 'Delete', 'Refresh', and 'Give feedback'. The 'Overview' tab is selected, showing a list of properties:

Property	Value
Resource group	TechBridge-Solutions-SA-TP
Location	South Africa North
Subscription	Azure subscription 1
Subscription ID	914ac7fd-7bfa-4f7a-9c7c-48a4630476f6
Address space	22.0.0.0/18
Subnets	2 subnets
DNS servers	Azure provided DNS service
BGP community stri...	Configure
Virtual network ID	9f821f1c-04ca-40a1-8e7f-c126aa36a602

Below the properties, there's a 'Tags' section with 'Add tags'. The 'Capabilities' tab is selected, showing a list of capabilities:

- DDoS protection: Configure additional protection from distributed denial of service attacks. Status: Not configured.
- Azure Firewall: Protect your network with a stateful L3-L7 firewall. Status: Not configured.
- Peering: Seamlessly connect two or more virtual networks. Status: Not configured.
- Microsoft Defender for Cloud: Strengthen the security posture of your environment.
- Private endpoints: Privately access Azure services without sending traffic across internet.

At the bottom, there's a taskbar with various application icons and a URL bar showing 'https://portal.azure.com/?utm_source=chatgpt.com#&@tristanpiek1972@gmail.com@microsoft.com'.

Microsoft Azure

portal.azure.com/?utm_source=chatgpt.com#@tristanpiek1972gmail.onmicrosoft.com/resource/subscriptions/914ac7fd-7bfa-4f7a-9c7c-48a4630476f6/resourceGroups/TechB...

Copilot

tristanpiek1972@gmail...
DEFAULT DIRECTORY (TRISTANPIE...

Create a resource

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Home > Network foundation | Virtual networks > TechBridge-Solutions-SA-VNet

TechBridge-Solutions-SA-VNet | Subnets

Virtual network

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Address space

Connected devices

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for Cloud

Network manager

DNS

Peering

Service endpoints

Add or remove favorites by pressing Ctrl+Shift+F

+ Subnet Refresh Manage users Delete Export to CSV

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table
☐	Public-Sub...	22.0.1.0/24	-	251	-	-	-
☐	Private-Su...	22.0.2.0/24	-	251	-	-	-

Showing 2 subnets

Give feedback

Microsoft Azure

portal.azure.com/?utm_source=chatgpt.com#@tristanpiek1972gmail.onmicrosoft.com/resource/subscriptions/914ac7fd-7bfa-4f7a-9c7c-48a4630476f6/resourceGroups/TechB...

Copilot

tristanpiek1972@gmail...
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Cost Management + Billing

Home > Network foundation | Network security groups

TechBridge-Solutions-NSG

Network security group

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Inbound security rules

Outbound security rules

Network interfaces

Subnets

Properties

Locks

Monitoring

Automation

Help

Add or remove favorites by pressing Ctrl+Shift+F

How do I create an alert to track firewall metric failures

Move Delete Refresh Give feedback

JSON View

Resource group (move) : TechBridge-Solutions-SA-TP

Custom security rules : 3 inbound, 0 outbound

Location : South Africa North

Associated with : 1 subnets, 0 network interfaces

Subscription (move) : Azure subscription 1

Subscription ID : 914ac7fd-7bfa-4f7a-9c7c-48a4630476f6

Tags (edit) : Add tags

Filter by name Port == all Protocol == all Source == all Destination == all Action == all

Priority ↑↓	Name ↑↓	Port ↑↓	Protocol ↑↓	Source ↑↓	Destination ↑↓	Action ↑↓
Inbound Security Rules						
100	Enable-HTTP	80	TCP	Any	Any	Allow
110	Enable-HTTPS	443	TCP	Any	Any	Allow
4000	Deny-Inbound	Any	Any	Any	Any	Deny
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalanc...	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound Security Rules						
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	Allow
65500	DenyAllOutBound	Any	Any	Any	Any	Deny

portal.azure.com/?utm_source=chatgpt.com#@tristanpiek1972gmail.onmicrosoft.com/resource/subscriptions/914ac7fd-7bfa-4f7a-9c7c-48a4630476f6/resourceGroups/TechB...Work

Microsoft AzureSearch resources, services, and docs (G+)

Copilot

tristanpiek1972@gmail...DEFAULT DIRECTORY (TRISTANPIE...

Create a resourceHomeDashboardAll servicesFAVORITESApplication TemplatesAll resourcesResource groupsApp ServicesFunction AppAzure SQL DatabaseAzure Cosmos DBVirtual machinesLoad balancersStorage accountsVirtual networksMicrosoft Entra IDMonitorAdvisorMicrosoft Defender for CloudCost Management + Billing

Home > Network foundation | Network security groups > TechBridge-Solutions-NSG

TechBridge-Solutions-NSG | SubnetsNetwork security group

Search resources, services, and docs (G+)

Search

Associate

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Outbound security rules

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Locks

Monitoring

Automation

Help

Search subnets

Name	Address range	Virtual network	
Public-Subnet	22.0.1.0/24	TechBridge-Solutions-SA-VNet	...

Give feedback

Add or remove favorites by pressing Ctrl+Shift+F